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BIOLOGICAL ANALYSIS AND MEDICAL IMAGING

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**High-sensitivity CRP:
an effective biomarker for predicting
cardiovascular risk**

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Abstract

According to the WHO, cardiovascular diseases (CVD) are the leading cause of death worldwide, particularly in Africa, which faces numerous care challenges such as weak diagnostic systems. It is therefore essential to establish an early and effective diagnosis for better management of these diseases. High-sensitivity C-reactive protein (hs-CRP) testing is a promising tool for cardiovascular risk assessment, particularly in the context of primary prevention. This biomarker is more effective for the early diagnosis of CVD than conventional markers such as total cholesterol. However, its interpretation must take into account certain inflammatory conditions that can raise its value.

Keywords: hs-CRP; cardiovascular diseases

Introduction

The prevalence of cardiovascular diseases (CVD) is rising worldwide. Their rapid progression in Africa, however, makes them a major public health problem, given the persistent difficulties in their management, such as limited diagnostic systems and delays in care. Consequently, integrating more effective biomarkers into early diagnosis, follow-up and management of at-risk patients would be a major asset in the fight against CVD, especially in Africa. New technologies and scientific advances have contributed to the identification of "new biomarkers" that appear more promising because they are more sensitive than conventional markers such as cholesterol. Among these new markers is high-sensitivity C-reactive protein, known as hs-CRP, which reveals the low-grade inflammation responsible for the development, progression and manifestation of cardiovascular diseases.

Role of hs-CRP in atherosclerosis and CVD risk assessment

Atherosclerosis is characterized by the deposition of lipid plaques on arterial walls, and inflammation is an integral part of its pathogenesis. A growing body of evidence shows that CRP is not merely a marker of inflammation, but that it also contributes to atherogenesis. Indeed, some studies show that hs-CRP may be involved in the progression of atherosclerosis through various mechanisms that are not yet fully understood. hs-CRP testing is therefore a promising tool for cardiovascular prevention, both primary and secondary. For example, studies have shown that a significant number of people who experienced cardiovascular events, or who died during a study, had "normal" cholesterol levels — suggesting that other markers such as hs-CRP may be essential for identifying risk.

Interpreting the assay

Specific hs-CRP levels are associated with different risk categories (Table):

- A level below 1 mg/L indicates a low risk of developing heart problems.
- A level above 3 mg/L is associated with a high risk.
- Values of interest range from 0.10 to 5.00 mg/L.

Cardiovascular risk	None	Low	High	Inflammation
hs-CRP level (mg/L)	< 1	1 – 3	3 – 6	> 6

- In a patient at moderate cardiovascular risk (SCORE between 1 and 5%), an hs-CRP > 3 mg/L may prompt the physician to reclassify the risk as high and intensify preventive management.

Challenges and limitations of hs-CRP use

Despite its promising potential, the use of the hs-CRP test is not without complexity or obstacles. Its precise measurement may be of no use for cardiovascular prevention if the patient has an infection (even minor, harmless bronchitis) or a chronic inflammatory disease. This underlines the importance of the clinical context when interpreting hs-CRP results.

Conclusion

While hs-CRP is a valuable biomarker for assessing CVD risk, its ability to detect the low-grade inflammation that lies at the heart of atherosclerosis makes it an important complement to cardiovascular risk assessment — particularly for identifying people who might otherwise be overlooked because their traditional markers, such as cholesterol levels, are normal. The use of hs-CRP as a routine diagnostic tool could considerably improve the management of CVD, especially in Africa. However, its interpretation must take into account acute or chronic inflammatory conditions that can raise CRP levels non-specifically.

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